Reproducibility Report for Study by Salomons et al. (2026, Animal Behaviour)

Reproduce Figure 2

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Table of contents

Load **1**

Reproducibility **5**
Individual Plots 5
Figure 2 20
Reproduce per-task models 25
Longitudinal only 25
Longitudinal + Control, like paper 27
Confirmatory mixed-effects logistic model 29
Effect Sizes (Longitudinal) 33

Data Simulation **36**

Load

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
```

```
v purrr     1.2.2
-- Conflicts ---------------------------------------- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(readxl)
install.packages("patchwork")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
  library(patchwork)
install.packages("lme4")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
  library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
auditory <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Auditory_Discr
```

```
Rows: 497 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
causal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Causal_Reasoning
```

```
Rows: 500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
detour <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Inhibit
```

```
Rows: 498 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
reversal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Revers
```

```
Rows: 496 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
marker <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Marker_Gesture.cs
```

```
Rows: 503 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
```

```
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
momentary <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Momentary_Poi
```

```
Rows: 495 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
odor <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Odor_Discriminatio
```

```
Rows: 492 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
unsolvable <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Unsolvable_E
```

```
Rows: 498 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (8): Dog ID, Age (wks), Age (days), Total Trials, Mean Look Time, # Tria...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
workingmem <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memo
```

```
Rows: 493 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
distraction <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Mem
```

```
Rows: 502 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Reproducibility

Individual Plots

```
auditory_long <- auditory |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

auditory_plot <- auditory_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
```

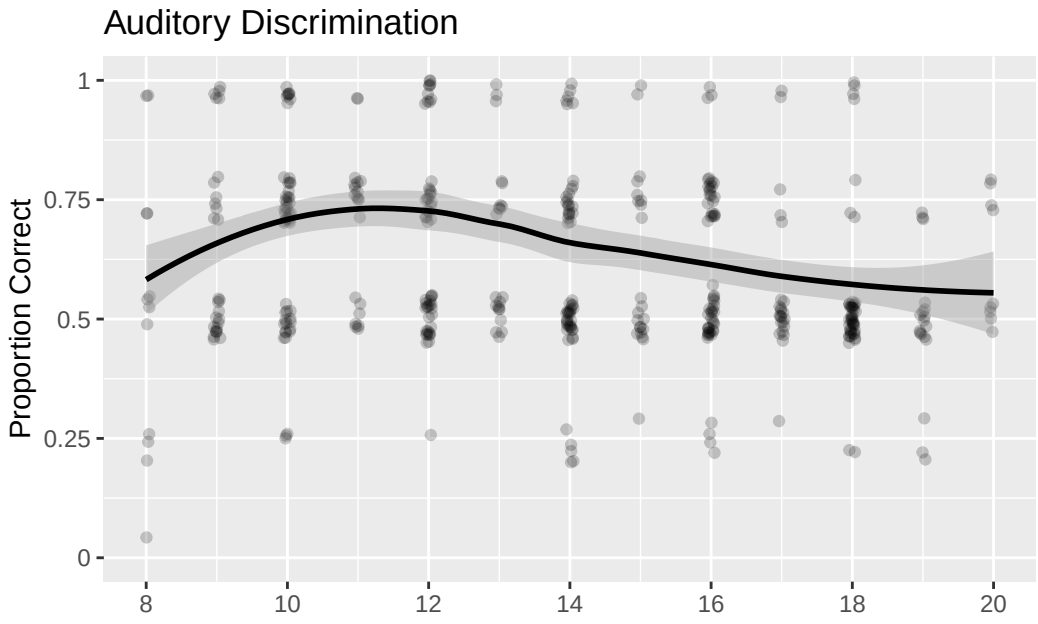
```r
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Auditory Discrimination")

print(auditory_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 64 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
causal_long <- causal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

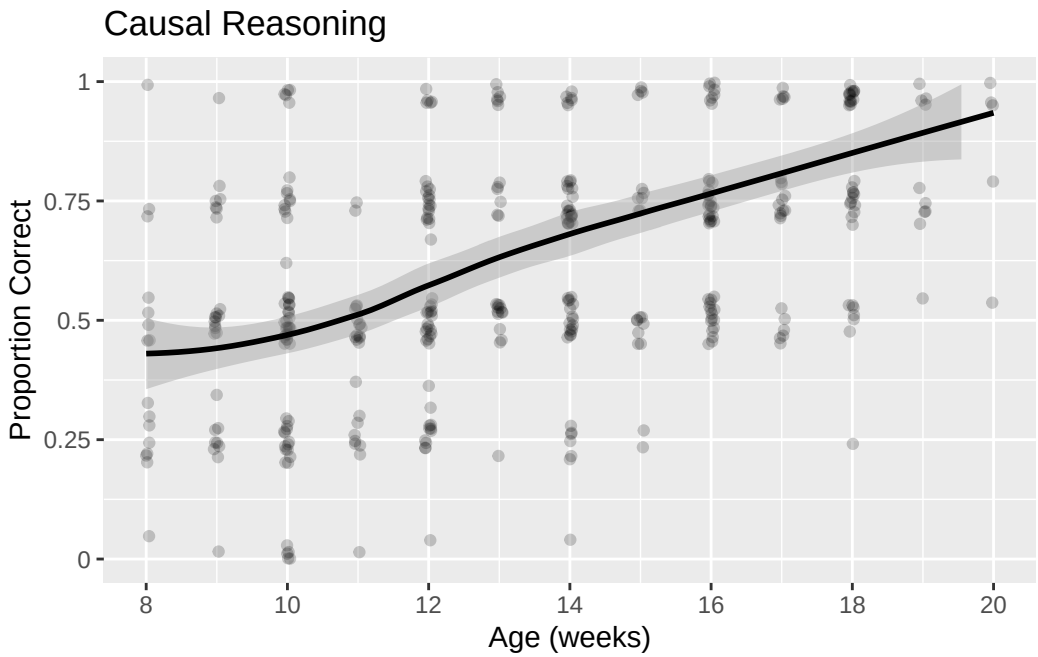
causal_plot <- causal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Causal Reasoning")

print(causal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 95 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
detour_long <- detour |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

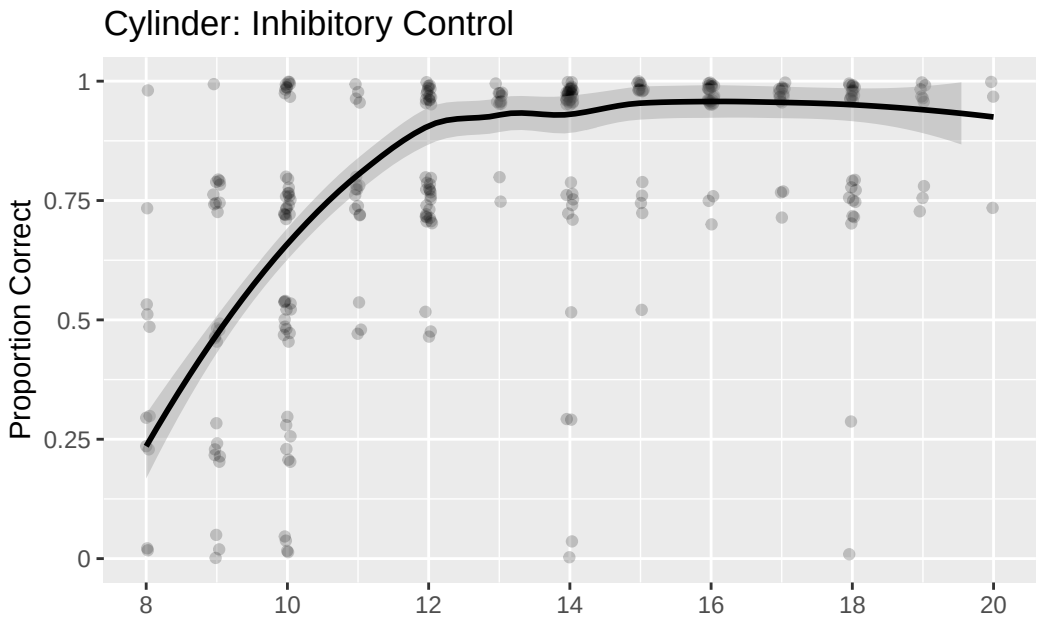
detour_plot <- detour_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Cylinder: Inhibitory Control")
```

```
print(detour_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 193 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
distraction_long <- distraction |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

distraction_plot <- distraction_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

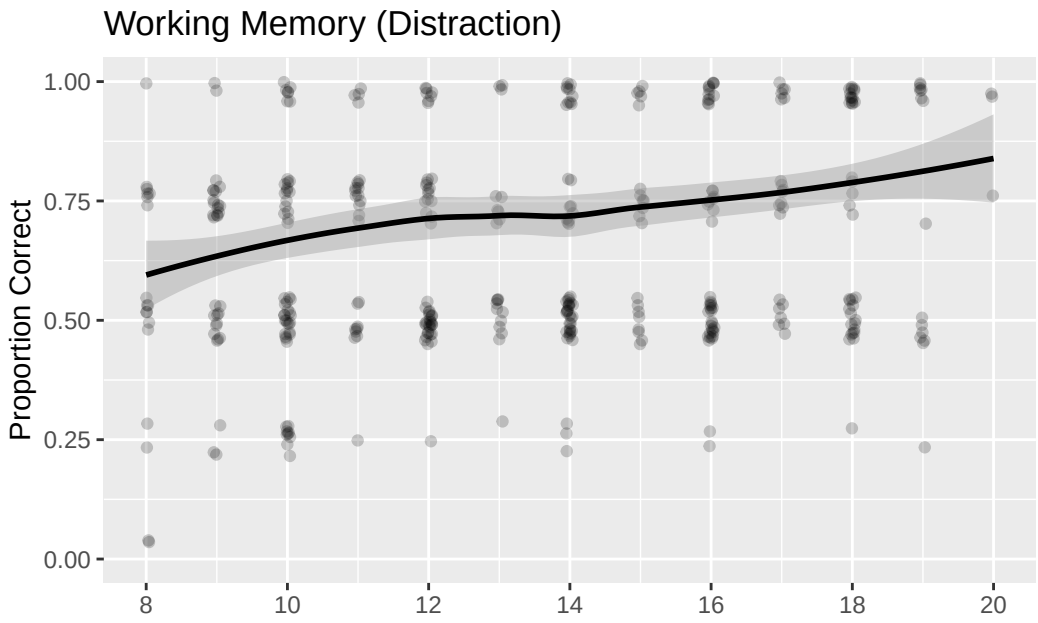
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2),
    labels = as.character
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Working Memory (Distraction)")

print(distraction_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 101 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
marker_long <- marker |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`),
      `experience` = as.numeric(`# Prior Test Sessions`)
  )

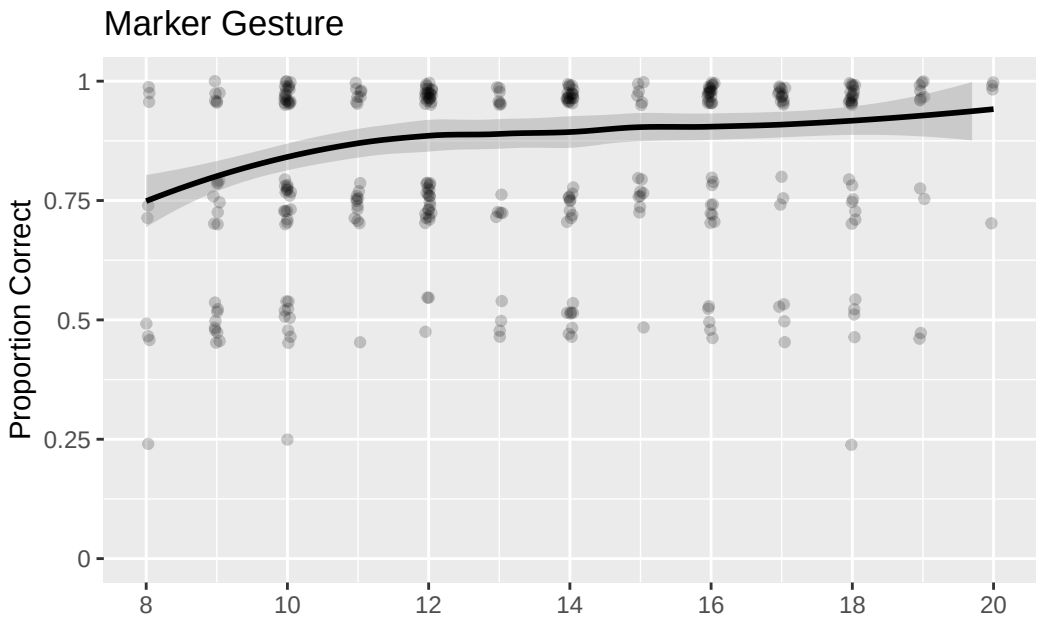
marker_plot <- marker_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Marker Gesture")
```

```
print(marker_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 168 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
momentary_long <- momentary |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

momentary_plot <- momentary_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

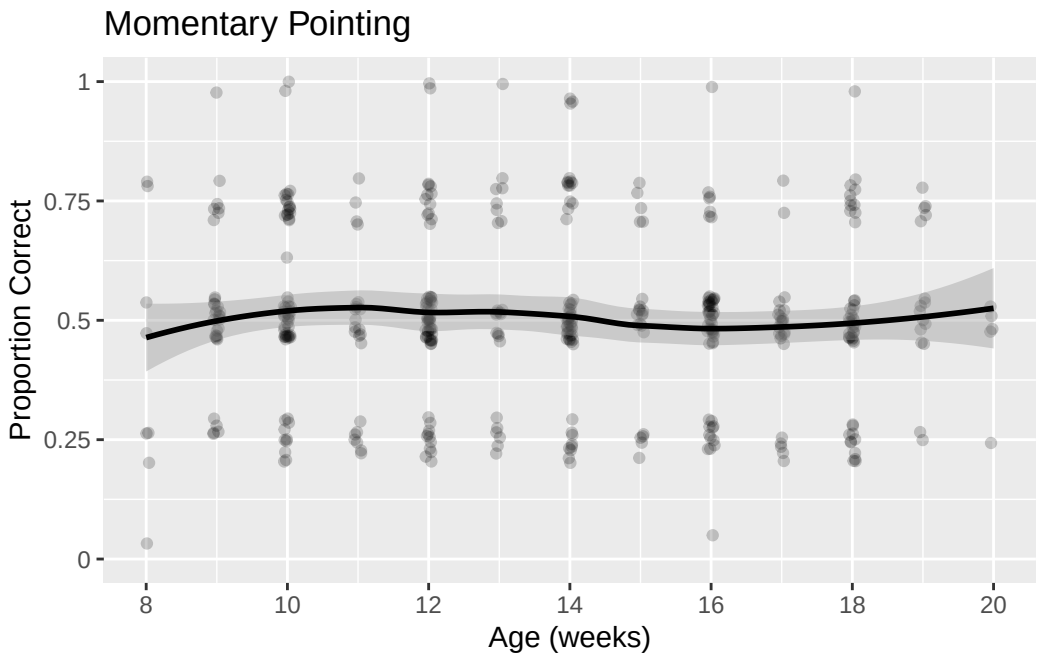
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Momentary Pointing")

print(momentary_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 47 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
odor_long <- odor |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`),
      `experience` = as.numeric(`# Prior Test Sessions`)
  )

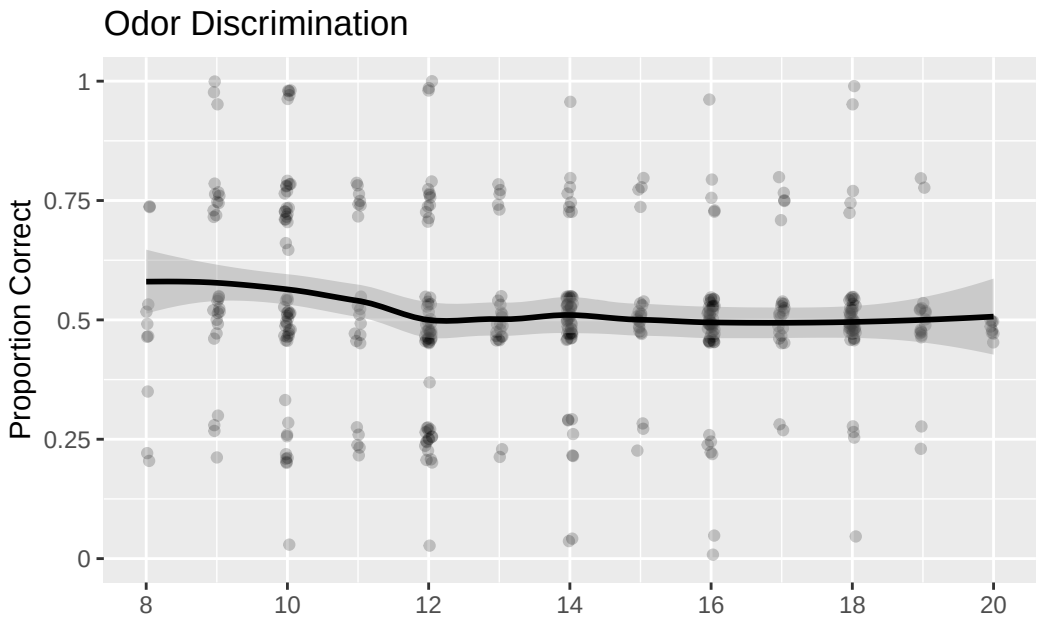
odor_plot <- odor_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Odor Discrimination")
```

```
print(odor_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 40 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
reversal_long <- reversal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

reversal_plot <- reversal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

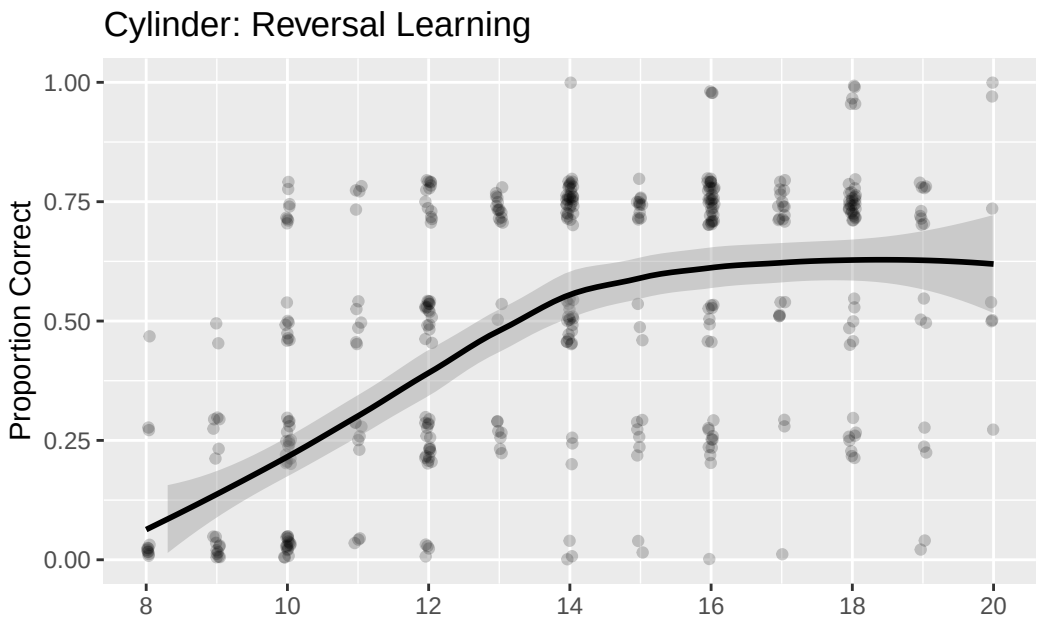
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Cylinder: Reversal Learning")

print(reversal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 77 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
unsolvable_long <- unsolvable |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Trials Looked` = as.numeric(`Proportion Trials Looked`),
    `Mean Look Time` = as.numeric(`Mean Look Time`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

proplooked_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Trials Looked`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Unsolvable (Proportion Looked)")

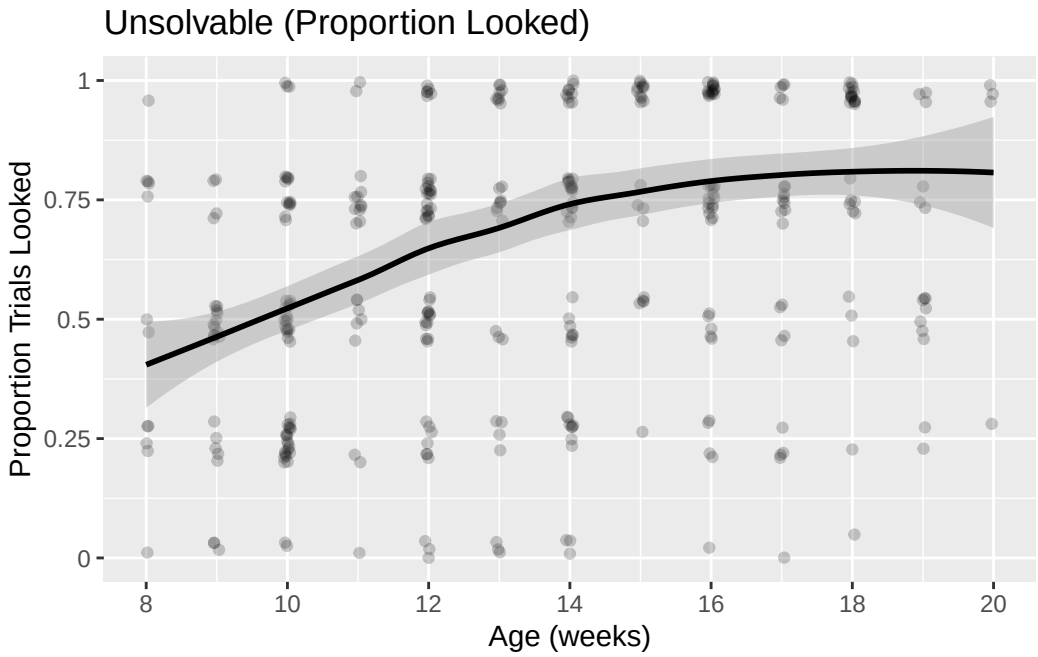
looktime_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Mean Look Time`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    name = "Mean Look Time (s)",
    limits = c(0, 20),
    breaks = seq(0, 20, by = 5)
  ) +
  labs(title = "Unsolvable (Looking Time)")

print(proplooked_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 122 rows containing missing values or values outside the scale range
(`geom_point()`).
```

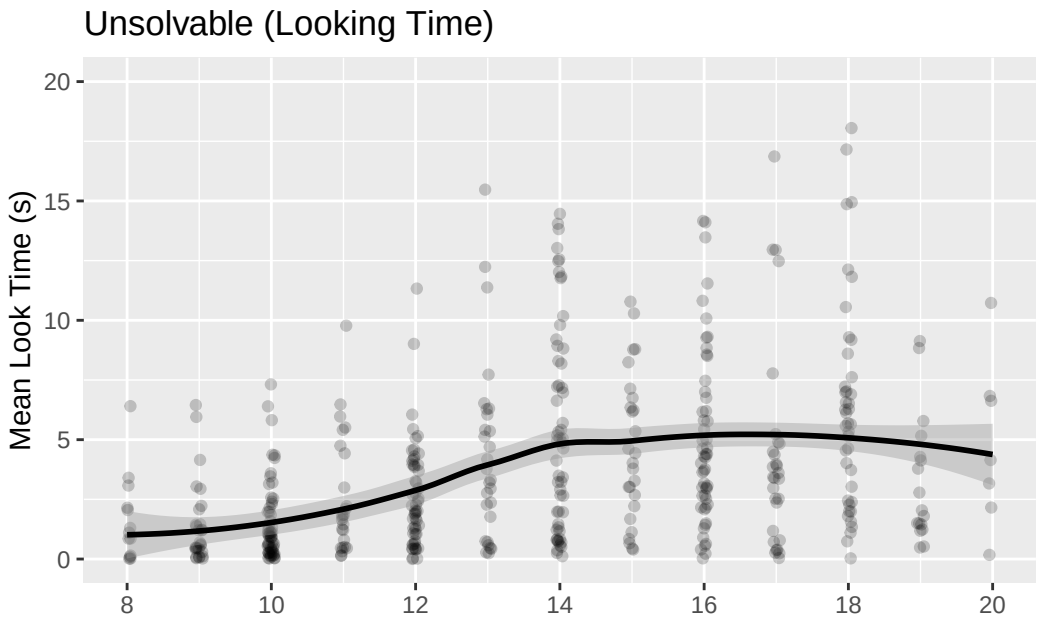



```
print(looktime_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 42 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
workingmem_long <- workingmem |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

workingmem_plot <- workingmem_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name ="",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Working Memory (Delay)")
```

```
print(workingmem_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 141 rows containing missing values or values outside the scale range
(`geom_point()`).
```

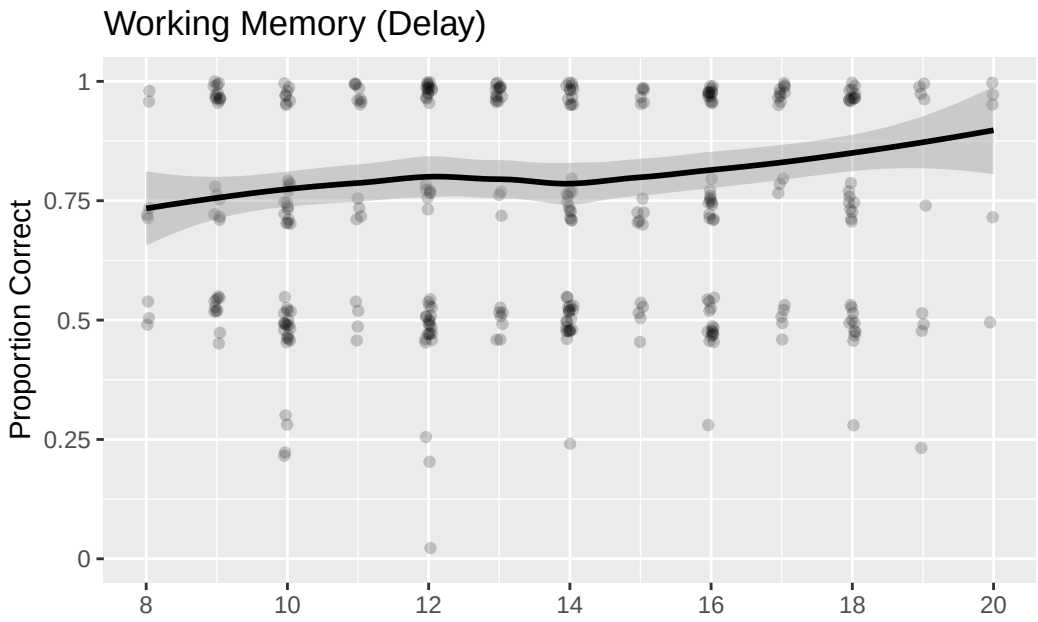


Figure 2

```
(marker_plot   | workingmem_plot | distraction_plot) /
(auditory_plot | odor_plot       | detour_plot)      /
(reversal_plot | looktime_plot   | proplooked_plot)  /
(causal_plot   | momentary_plot  | plot_spacer())&
  theme_grey(base_family = "serif") + #match theme to original
  theme(
```

```
    plot.title = element_text(hjust = 0.5, size = 11),
    axis.line  = element_line(colour = "black"),
    axis.title = element_text(size = 10),
    axis.text  = element_text(size = 9, colour = "black")
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 167 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 125 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 101 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 71 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 36 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 199 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 66 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 35 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 117 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

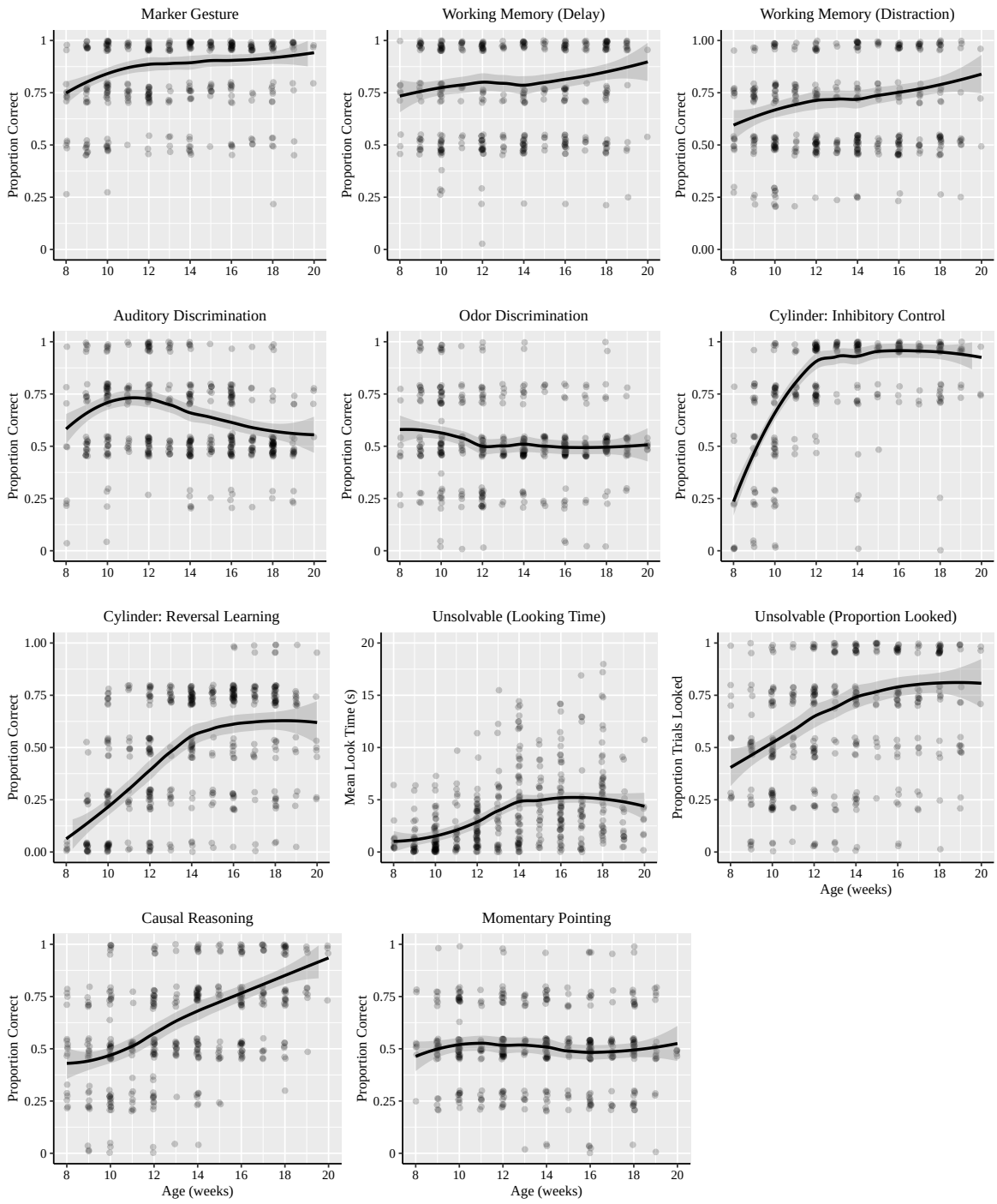
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 97 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 39 rows containing missing values or values outside the scale range
(`geom_point()`).
```


Marker Gesture
Working Memory (Delay)
Working Memory (Distraction)
Auditory Discrimination
Odor Discrimination
Cylinder: Inhibitory Control
Cylinder: Reversal Learning
Unsolvable (Looking Time)
Unsolvable (Proportion Looked)
Causal Reasoning
Momentary Pointing
Proportion Correct
Mean Look Time (s)
Proportion Trials Looked
Age (weeks)

Reproduce per-task models

Longitudinal only

```
task_models <- function(df) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                 family = binomial, data = df)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + exp
                 family = binomial, data = df)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble <- bind_rows(
  "Marker gesture"               = task_models(marker_long),
  "Working memory (Delay)"       = task_models(workingmem_long),
  "Working memory (Distraction)" = task_models(distraction_long),
  "Auditory discrimination"      = task_models(auditory_long),
  "Odor discrimination"         = task_models(odor_long),
  "Cylinder: inhibitory control" = task_models(detour_long),
  "Cylinder: reversal learning"  = task_models(reversal_long),
  "Causal reasoning"             = task_models(causal_long),
  "Momentary pointing"           = task_models(momentary_long),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   844.        846. age
2 Working memory (Delay)          1050.       1048. age + experience
3 Working memory (Distraction)    1168.       1169. age
```

```
4 Auditory discrimination        1226.       1221. age + experience
5 Odor discrimination            1167.       1169. age
6 Cylinder: inhibitory control    938.        895. age + experience
7 Cylinder: reversal learning    1312.       1298. age + experience
8 Causal reasoning               1210.       1186. age + experience
9 Momentary pointing             1219.       1219. age + experience
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable_long)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` + e
               family = binomial, data = unsolvable_long)

aic_tibble2 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble2
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1352.       1353. age
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + experience + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)


aic_tibble3 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble3
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2514.       2515. age
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 844.3)

- Working Memory with Distraction (AIC = 1168.1)
- Odor Discrimination (AIC = 1166.9)
- Unsolvable Proportion Looked (AIC = 1351.6)
- Unsolvable Average Looking Time (AIC = 2514.5).

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1047.9)
- Auditory Discrimination (AIC = 1220.9)
- Cylinder Inhibitory Control (AIC = 894.9)
- Cylinder Reversal Learning (AIC = 1297.8)
- Causal Reasoning (AIC = 1185.8).
- Momentary Pointing (AIC = 1218.8).

Longitudinal + Control, like paper

```
task_models2 <- function(df2) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                  family = binomial, data = df2)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + `#
                  family = binomial, data = df2)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble4 <- bind_rows(
  "Marker gesture"               = task_models2(marker),
  "Working memory (Delay)"       = task_models2(workingmem),
  "Working memory (Distraction)" = task_models2(distraction),
  "Auditory discrimination"      = task_models2(auditory),
  "Odor discrimination"         = task_models2(odor),
  "Cylinder: inhibitory control" = task_models2(detour),
  "Cylinder: reversal learning"  = task_models2(reversal),
  "Causal reasoning"             = task_models2(causal),
  "Momentary pointing"           = task_models2(momentary),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble4
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   876.        878. age
2 Working memory (Delay)          1081.       1073. age + experience
3 Working memory (Distraction)    1223.       1224. age
4 Auditory discrimination         1285.       1277. age + experience
5 Odor discrimination             1226.       1227. age
6 Cylinder: inhibitory control    1089.        957. age + experience
7 Cylinder: reversal learning     1427.       1364. age + experience
8 Causal reasoning                1327.       1244. age + experience
9 Momentary pointing              1269.       1270. age
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)

aic_tibble5 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble5
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1443.       1431. age + experience
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + `# Prior Test Sessions` + (1 | `Dog ID`),
```

```
              data = unsolvable, REML = FALSE)


aic_tibble6 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble6
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2640.       2629. age + experience
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 876.1)
- Working Memory with Distraction (AIC = 1223.1)
- Odor Discrimination (AIC = 1225.8)
- Momentary Pointing (AIC =)

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1072.8)
- Auditory Discrimination (AIC = 1276.8)
- Cylinder Inhibitory Control (AIC = 957.2)
- Cylinder Reversal Learning (AIC = 1364.4)
- Causal Reasoning (AIC = 1244.3).
- Unsolvable Proportion Looked (AIC = 1431.1)
- Unsolvable Average Looking Time (AIC = 2629.2).

Confirmatory mixed-effects logistic model

```r
all_tasks <- bind_rows(
  "Marker gesture"                = marker_long,
  "Working memory (Delay)"        = workingmem_long,
  "Working memory (Distraction)" = distraction_long,
  "Auditory discrimination"       = auditory_long,
  "Odor discrimination"           = odor_long,
  "Cylinder: inhibitory control" = detour_long,
  "Cylinder: reversal learning"  = reversal_long,
  "Causal reasoning"              = causal_long,
  "Momentary pointing"            = momentary_long,
  .id = "task") |>
  transmute(task,
            id      = factor(`Dog ID`),
            age     = `Age (wks)` - 8,
            correct = `Trials Correct`,
            total   = `Total Trials`)

ctrl <- glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))

r_age_exp <- bind_rows(marker_long, workingmem_long, distraction_long, auditory_long,
                       odor_long, detour_long, reversal_long, causal_long,
                       momentary_long, unsolvable_long) |>
  summarise(r = cor(`Age (wks)`, `# Prior Test Sessions`)) |>
  pull(r)
```

```r
# random intercepts only
m1 <- glmer(cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

# by-puppy random slope for age
m2 <- glmer(cbind(correct, total - correct) ~ age * task + (1 + age | id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

isSingular(m2)   # TRUE = random effects too complex for the data
```

```
[1] TRUE
```

```r
anova(m1, m2)    # does the random slope improve fit?
```

```
Data: all_tasks
Models:
```

```
m1: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
m2: cbind(correct, total - correct) ~ age * task + (1 + age | id) + (1 | id:task)
   npar   AIC   BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
m1   20 10244 10371 -5102.0     10204
m2   22 10251 10391 -5103.4     10207     0  2          1
```

```
# same slope as m2, no intercept-slope correlation
m3 <- glmer(cbind(correct, total - correct) ~ age * task + (1 + age || id) + (1 | id:task),
            family = binomial, data = all_tasks, control = ctrl)

isSingular(m3)    # FALSE = fitted cleanly
```

```
[1] FALSE
```

```
anova(m1, m3)     # no sig improvement, m1 has slightly better AIC
```

```
Data: all_tasks
Models:
m1: cbind(correct, total - correct) ~ age * task + (1 | id) + (1 | id:task)
m3: cbind(correct, total - correct) ~ age * task + ((1 | id) + (0 + age | id)) + (1 | id:tas
   npar   AIC   BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
m1   20 10244 10371 -5102.0     10204
m3   21 10246 10379 -5101.8     10204 0.4436  1     0.5054
```

```
# m3 most complex model that converged without warnings
m_final <- m3

# KEY RESULT: age x task interaction
m_final_noint <- update(m_final, . ~ . - age:task)
lrt_m3 <-anova(m_final_noint, m_final)

# m1 has marginally lower AIC so using as robustness check
m_final2 <- m1

# age x task interaction
m_final2_noint <- update(m_final2, . ~ . - age:task)
lrt_m1 <- anova(m_final2_noint, m_final2)
```

```
all_tasks_all <- bind_rows(
  "Marker gesture"               = marker,
```

```
  "Working memory (Delay)"       = workingmem,
  "Working memory (Distraction)" = distraction,
  "Auditory discrimination"      = auditory,
  "Odor discrimination"          = odor,
  "Cylinder: inhibitory control" = detour,
  "Cylinder: reversal learning"  = reversal,
  "Causal reasoning"             = causal,
  "Momentary pointing"           = momentary,
  .id = "task") |>
  transmute(task,
            id         = factor(`Dog ID`),
            age        = `Age (wks)` - 8,
            experience = `# Prior Test Sessions`,
            correct    = `Trials Correct`,
            total      = `Total Trials`)

cor(all_tasks_all$age, all_tasks_all$experience)   #0.76
```

```
[1] 0.7659694
```

```
# Each task gets its own age slope AND its own experience slope
m_ae <- glmer(cbind(correct, total - correct) ~ age * task + experience * task +
                (1 | id) + (1 | id:task),
              family = binomial, data = all_tasks_all, control = ctrl)

# Does age x task matter AFTER accounting for task-specific experience effects?
m_ae_noint <- update(m_ae, . ~ . - age:task)
lrt_ae <- anova(m_ae_noint, m_ae)
lrt_ae
```

```
Data: all_tasks_all
Models:
m_ae_noint: cbind(correct, total - correct) ~ age + task + experience + (1 | id) + (1 | id:ta
m_ae: cbind(correct, total - correct) ~ age * task + experience * task + (1 | id) + (1 | id:t
           npar   AIC   BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
m_ae_noint   21 10658 10792 -5307.9     10616
m_ae         29 10630 10816 -5286.3     10572 43.33  8  7.613e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The age × task interaction was significant in the random-slope model (m3), χ^2(8) = 633.17, p <0.001, and in the random-intercepts model (m1), χ^2 (8) = 632.76, p <0.001.

This indicates that the 9 tasks do not share one developmental rate, consistent with the multiple intelligences hypothesis. Because age and experience are highly correlated in the longitudinal puppies (r = 0.91), this model cannot disentangle the model whether these task differences reflect maturation, accumulated testing experience, or both.

After adding the 22 control puppies and task-specific experience effects, the age × task interaction was still significant, χ^2 (8) = 43.33, p <0.001. While part of the longitudinal age × task interaction effect may be shared with experience , the results from the 113-subject sample further support the multiple intelligences hypothesis, although age and experience remain partly correlated (r = 0.77).

Effect Sizes (Longitudinal)

```
install.packages("emmeans")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
# slopes on the probability scale
slopes <- emtrends(m_final, ~ task, var = "age")
slopes_table <- as_tibble(slopes) |>
  mutate(OR_per_week = exp(age.trend))
slopes_table
```

```
# A tibble: 9 x 7
  task                    age.trend     SE    df asymp.LCL asymp.UCL OR_per_week
  <fct>                       <dbl>  <dbl> <dbl>     <dbl>     <dbl>       <dbl>
1 Auditory discrimination  -0.0599  0.0151   Inf   -0.0896  -0.0302        0.942
2 Causal reasoning          0.231   0.0171   Inf    0.198    0.265         1.26
3 Cylinder: inhibitory c~   0.466   0.0292   Inf    0.409    0.523         1.59
4 Cylinder: reversal lea~   0.254   0.0167   Inf    0.222    0.287         1.29
5 Marker gesture            0.118   0.0227   Inf    0.0736   0.162         1.13
6 Momentary pointing       -0.00658 0.0144   Inf   -0.0348   0.0217        0.993
```

```
7 Odor discrimination      -0.0362  0.0145   Inf   -0.0647  -0.00777       0.964
8 Working memory (Delay)    0.0607  0.0186   Inf    0.0244   0.0971        1.06
9 Working memory (Distra~   0.0883  0.0164   Inf    0.0563   0.120         1.09
```

```
# age slope per task on the log-odds scale
slopes_prob <- emtrends(m_final, ~ task, var = "age", regrid = "response")
slopes_prob
```

```
 task                          age.trend      SE  df asymp.LCL asymp.UCL
 Auditory discrimination       -0.01343 0.00340 Inf  -0.02008  -0.00677
 Causal reasoning               0.05056 0.00373 Inf   0.04325   0.05787
 Cylinder: inhibitory control   0.04058 0.00281 Inf   0.03507   0.04608
 Cylinder: reversal learning    0.06237 0.00410 Inf   0.05433   0.07041
 Marker gesture                 0.01124 0.00210 Inf   0.00713   0.01534
 Momentary pointing            -0.00164 0.00360 Inf  -0.00870   0.00541
 Odor discrimination           -0.00904 0.00362 Inf  -0.01614  -0.00194
 Working memory (Delay)         0.00905 0.00276 Inf   0.00363   0.01446
 Working memory (Distraction)   0.01716 0.00316 Inf   0.01097   0.02335

Confidence level used: 0.95
```

```
prob_table <- as_tibble(slopes_prob) |>
  select(task, prob_per_week = age.trend)

effect_table <- as_tibble(slopes) |>
  transmute(task,
            OR      = exp(age.trend),
            OR_low  = exp(asymp.LCL),
            OR_high = exp(asymp.UCL)) |>
  left_join(prob_table, by = "task") |>
  arrange(desc(OR)) |>
  transmute(Task = task,
            `OR per week [95% CI]` = sprintf("%.2f [%.2f, %.2f]", OR, OR_low, OR_high),
            `Change in proportion correct per week` = sprintf("%+.3f", prob_per_week))
effect_table
```

```
# A tibble: 9 x 3
  Task                          `OR per week [95% CI]` Change in proportion cor~1
  <fct>                         <chr>                  <chr>
1 Cylinder: inhibitory control 1.59 [1.51, 1.69]      +0.041
2 Cylinder: reversal learning  1.29 [1.25, 1.33]      +0.062
```

```
3 Causal reasoning             1.26 [1.22, 1.30]      +0.051
4 Marker gesture               1.13 [1.08, 1.18]      +0.011
5 Working memory (Distraction) 1.09 [1.06, 1.13]      +0.017
6 Working memory (Delay)       1.06 [1.02, 1.10]      +0.009
7 Momentary pointing           0.99 [0.97, 1.02]      -0.002
8 Odor discrimination          0.96 [0.94, 0.99]      -0.009
9 Auditory discrimination      0.94 [0.91, 0.97]      -0.013
# i abbreviated name: 1: `Change in proportion correct per week`
```

```
# pairwise comparisons between tasks (Tukey-adjusted)
between_tasks <- as_tibble(pairs(slopes)) |>
  mutate(ratio_of_ORs = exp(estimate))
between_tasks
```

```
# A tibble: 36 x 7
   contrast                  estimate     SE    df z.ratio  p.value ratio_of_ORs
   <chr>                        <dbl>  <dbl> <dbl>   <dbl>    <dbl>        <dbl>
 1 Auditory discrimination ~  -0.291  0.0227   Inf -12.8   0                0.748
 2 Auditory discrimination ~  -0.526  0.0328   Inf -16.1   0                0.591
 3 Auditory discrimination ~  -0.314  0.0224   Inf -14.0   0                0.730
 4 Auditory discrimination ~  -0.178  0.0272   Inf  -6.55  2.05e- 9        0.837
 5 Auditory discrimination ~  -0.0533 0.0208   Inf  -2.57  2.00e- 1        0.948
 6 Auditory discrimination ~  -0.0237 0.0208   Inf  -1.13  9.69e- 1        0.977
 7 Auditory discrimination ~  -0.121  0.0238   Inf  -5.06  1.46e- 5        0.886
 8 Auditory discrimination ~  -0.148  0.0222   Inf  -6.69  8.11e-10        0.862
 9 Causal reasoning - Cylin~  -0.235  0.0337   Inf  -6.97  1.11e-10        0.791
10 Causal reasoning - Cylin~  -0.0231 0.0237   Inf  -0.976 9.88e- 1        0.977
# i 26 more rows
```

```
between_tasks_sig <- between_tasks |>
  mutate(sig = p.value<0.05) |>
  count(sig == "TRUE")
between_tasks_sig
```

```
# A tibble: 2 x 2
  `sig == "TRUE"`     n
  <lgl>           <int>
1 FALSE               8
2 TRUE               28
```

Each task's age slope is reported as an odds ratio (OR) per week. Effects were largest for cylinder inhibitory control (OR [95% CI] = 1.59 [1.51, 1.69]), cylinder reversal learning (OR

= 1.29 [1.25, 1.33]), and causal reasoning (OR = 1.26 [1.22, 1.30]); smaller for marker gesture (OR = 1.13 [1.08, 1.18]), working memory with distraction (OR = 1.09 [1.06, 1.13]), and working memory with delay (OR = 1.06 [1.02, 1.10]); absent for momentary pointing (OR = 0.99 [0.97, 1.02]); and slightly negative for auditory discrimination (OR = 0.94 [0.91, 0.97]) and odor discrimination (OR = 0.96 [0.94, 0.99]).

When using a probability scale to look at proportion correct by age, cylinder inhibitory control increased by about 4 percentage points per week, less than cylinder reversal learning (about 6 points per week), likely because scores already neared cieling by mid-testing. Comparing tasks directly showed that most developed at significantly different rates (significant for 28 of 36 comparisons), again reflecting the multiple intelligences hypothesis. For example, cylinder inhibitory control improved 1.65 times faster per week than odor discrimination ($p < .001$), whereas causal reasoning and reversal learning developed at similar rates (ratio = 0.98, $p = 0.99$).

Data Simulation